

12. What is the difference between linking a library statically with your application, versus linking it dynamically? Which is preferred, a statically linked library or a dynamically linked library? Justify your answer.
13. Let's suppose that you have written an application, and it has been working perfectly on a 32-bit platform. Now you are asked to port your application to a 64-bit platform. What kind of coding changes would be needed?
14. You have written an application, and it has been working well on a little-endian platform. Now you are asked to port your application to a big-endian platform. Besides, your application also reads from files stored on disks, processes the data, and writes back data to disk in output files. What, if any, kind of coding changes would be needed?
15. We all have heard of paging and virtual memory. Does the operating system maintain a page table per process, or is there only one page table for all processes in the system? Explain your answer.
16. In a multiprocessor system with each processor having its own cache, how are the updates made by one processor to a location in its cache made visible to other processors? Does this require hardware support? If not, can you sketch what kind of software support would be needed from the operating system?
17. Given two threads running on two different processors, Thread A running on processor 1 updates the global variable X first, and then the global variable Y . Can thread B running on processor 2 see the updated value of Y before it sees the updated value of X ? Explain your answer.
18. You are given three arrays A , B and C , each containing N distinct integers. You are asked to find out whether there exist three such integers $a \in A$, $b \in B$ and $c \in C$ such that $a + b + c = 0$. What is the complexity of your solution? Would the complexity be the same if you are asked to repeat this problem for the two arrays, A and B only, such that there exist any two integers $a \in A$ and $b \in B$ for which $a + b = 0$?
19. Can you write to a memory location that is not present in the processor cache? If yes, how? If not, why not?
20. We all have used debuggers, and we know that we can set breakpoints to stop program execution at a particular program location. Can you explain how breakpoints are implemented? Does it require underlying hardware support?
21. What is meant by privileged instructions? In a virtualised system (system virtual machine, such as Xen/KVM/Hyper V), how are privileged instructions handled?
22. Given an array of N integers, which is mostly sorted, with very few elements being out of sort (the number of unsorted elements is k , where $k \ll N$), what sorting algorithm would you use to sort the array, and why?
23. Given a 32-bit representation of an integer, write a code snippet which, given a bit position k , prints the current bit value at that location, and then flips it.

24. What is the problem with the following code snippet?

```
int* bar(int size)
{
    int* p = alloca(size);
    return p;
}

void foo(int n)
{
    int* myspace = bar(n);
    *myspace = 10;
    ...
}
```

My 'must read book' for this month

This month, our reader Srividya K, recommends the book '*Concurrent Programming in Java*' by Doug Lea. Srividya says that this book helped him understand and apply concurrent programming techniques effectively in developing software for commercial projects. He also cautions readers that this is not a book for beginners to Java. While on the subject of concurrency, there is an excellent article that discusses how efficiently a lock-free concurrent hash table can be implemented. It is available at: http://www.azulsystems.com/events/javaone_2007/2007_LockFreeHash.pdf. This is a good read not just for Java programmers, but for anyone writing concurrent code. It is also worth reading Cliff Click's presentation on writing scalable non-blocking code for Java at: http://www.azulsystems.com/events/javaone_2008/2008_CodingNonBlock.pdf. The code corresponding to the data structures discussed in his article is available on Sourceforge at <http://sourceforge.net/projects/high-scale-lib/>.

If you have a favourite programming book that you think is a must-read for every programmer, please send me a note with the book's name, and a short note on why you think it is useful. I will mention it in this column. This will help many readers who want to improve their coding skills.

Please send me your answers to this month's questions. If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming and wishing you the very best! 

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The author is a specialist in compiler optimisation, and works at Hewlett-Packard India. She has a number of publications and patents to her credit, and her areas of interest include virtualisation technologies and software development tools.

[If you are preparing for computer science programming interviews, you may find it useful to visit Sandya's programming interviews discussion group 'Computer Science Interview Training (India)' on LinkedIn (www.linkedin.com), and her website <http://www.tinterviewtraining.com>.]